1 (a)	$254 = 2 + 3(n-1) \Rightarrow n = ...$	M1
	85	A1
		(2)
(b)	$\frac{"85"}{2}\left(2\times 2 + ("85"-1)\times 3\right)$	M1
	10 880	A1
		(2)
		4 marks

2 (a)	$(2-5x)^8 = 2^8 + {}^8C_1(2)^7(-5x)^1 + {}^8C_2(2)^6(-5x)^2 + {}^8C_3(2)^5(-5x)^3 + \ldots$	
	Constant term 2^8 OR 256	B1
	Correct form for terms 3 or 4: ${}^8C_2(2)^6(\pm 5x)^2$ or ${}^8C_3(2)^5(\pm 5x)^3$	M1
	Correct un-simplified expansion for 2 out of 3 terms from terms 2, 3 and 4: $\ldots + 8(2)^7(-5x)^1 + 28(2)^6(-5x)^2 + 56(2)^5(-5x)^3 + \ldots$	A1
	Correct simplified expansion $256 - 5120x + 44800x^2 - 224000x^3 + \ldots$	A1
		(4)
(b)	-0.01	B1
		(1)
		(5 marks)

	OPEN TOPPED	CLOSED CONTAINER	
3.(a)	$120 = 3x^2y \Rightarrow y = \dfrac{40}{x^2}$	$120 = 3x^2y \Rightarrow y = \dfrac{40}{x^2}$	M1, A1
	$A = 3x^2 + 8xy$ $= 3x^2 + 8x \times \dfrac{40}{x^2} = 3x^2 + \dfrac{320}{x}$	$A = 6x^2 + 8xy$ $= 6x^2 + 8x \times \dfrac{40}{x^2} = 6x^2 + \dfrac{320}{x}$	dM1, A1 **(4)**
(b)	$\dfrac{dA}{dx} = 6x - \dfrac{320}{x^2}$	$\dfrac{dA}{dx} = 12x - \dfrac{320}{x^2}$	M1, A1ft
	$\dfrac{dA}{dx} = 6x - \dfrac{320}{x^2} = 0$ $\Rightarrow x^3 = \dfrac{160}{3} \Rightarrow x =$ awrt 3.76	$\dfrac{dA}{dx} = 12x - \dfrac{320}{x^2} = 0$ $\Rightarrow x^3 = \dfrac{80}{3} \Rightarrow x =$ awrt 2.99	dM1, A1 **(4)**
(c)	Attempts $\left.\dfrac{d^2A}{dx^2}\right\|_{x=3.76} = 6 + \dfrac{640}{x^3}$ $= 6 + \dfrac{640}{3.76^3} = ..$	Attempts $\left.\dfrac{d^2A}{dx^2}\right\|_{x=2.99} = 12 + \dfrac{640}{x^3}$ $= 12 + \dfrac{640}{2.99^3} = ..$	M1
	$\left.\dfrac{d^2A}{dx^2}\right\|_{x=3.76}$ = awrt 18 > 0 MINIMUM*	$\left.\dfrac{d^2A}{dx^2}\right\|_{x=2.99}$ = awrt 36 > 0 MINIMUM*	A1* **(2)**
			(10 marks)

4 (a)	$\frac{a\left(1-r^3\right)}{1-r}=70.2\ ,\ \frac{a}{1-r}=75$	B1, B1
	Sub (2) in (1) $\quad 75\left(1-r^3\right)=70.2\Rightarrow r^3=\left(0.064\right)$	M1
	$r=0.4$	A1
		(4)
(b)	Substitutes $r=0.4$ into $\frac{a}{1-r}=75$ and finds a	M1
	$a=45$	A1
		(2)
		(6 marks)

5 (a)	$\mathrm{f}(x) = 3x^3 + ax^2 - 10x + b$	
	$\mathrm{f}\left(\frac{4}{3}\right) = 0 \Rightarrow 3\times\left(\frac{4}{3}\right)^3 + a\left(\frac{4}{3}\right)^2 - 10\left(\frac{4}{3}\right) + b = 0$	M1
	$\frac{64}{9} + \frac{16}{9}a - \frac{40}{3} + b = 0 \Rightarrow 64 + 16a - 120 + 9b = 0 \quad \Rightarrow 16a + 9b = 56$ *	A1*
		(2)
(b)	$\mathrm{f}(2) = b \Rightarrow 3\times(2)^3 + a(2)^2 - 10(2) + b = b$ leading to an equation in just a	M1
	$4a = -4 \Rightarrow a = -1$	A1
	Substitutes $a = '-1'$ in $-16 + 9b = 56 \Rightarrow b = ...$	M1
	$\Rightarrow b = 8$	A1
		(4)
(c)	$\mathrm{f}(x) = 3x^3 - x^2 - 10x + 8 = (3x-4)(x^2 + x - 2)$	M1, A1
	$= (3x-4)(x+2)(x-1)$	A1
		(3)
		(9 marks)

6(a)	States or uses $M = (5, 4)$ **or** grad $AB = \frac{1}{3}$ o.e.	B1
	States or uses $M = (5, 4)$ **and** grad $AB = \frac{1}{3}$ o.e.	B1
	Equation of l is $y - 4 = -3(x - 5)$	M1
	$y = -3x + 19$	A1
		(4)
(b)	$k = -2$	B1 ft
		(1)
(c)	Attempts radius or radius ^{2} E.g. $(11 - 7)^2 + (6 - -2)^2$	M1
	States $(x - 7)^2 + (y + '2')^2 = (11 - 7)^2 + (6 - -2)^2$	dM1
	$(x - 7)^2 + (y + 2)^2 = 80$	A1
		(3)
		(8 marks)

7 (i)	$\frac{1}{2}\{\log 4+\log 9+2(\log 5+\log 6+\log 7+\log 8)\}$	B1
	$=\frac{1}{2}\log(4\times 9)+\log(5\times 6\times 7\times 8) \quad =\log 10080$	M1 A1
		(3)
(ii)	$2\log_5(5-a)-\log_5(a+25)=1$	
	$2\log_5(5-a)=\log_5(5-a)^2$ **OR** $1=\log_5 5$	B1
	$\log_5\frac{(5-a)^2}{(a+25)}=1$	M1
	$a^2-15a-100=0$	A1
	$(a-20)(a+5)=0$	dM1
	$a=-5$	A1
		(5)
		(8 marks)

8(i)	Provides a counter example E.g. $\sqrt{2}\times\sqrt{8}=4$	B1 **(1)**
(ii)	Knows that odd numbers are of the form $2k+1, \quad (k\in\mathbb{N})$	B1
	Attempts $(2k+1)^3+3(2k+1)+2=....$	M1
	$=8k^3+12k^2+12k+6$	A1
	$=4\times(2k^3+3k^2+3k+1)+2$ Which is even but not a multiple of 4. Hence proven	A1* **(4)** **(5 marks)**

	$y = \dfrac{9x^2\left(5-\sqrt{x}\right)}{5}$	
9(a)	$y = 9x^2 - \dfrac{9}{5}x^{\frac{5}{2}} \Rightarrow \dfrac{dy}{dx} = 18x - \dfrac{9}{2}x^{\frac{3}{2}}$	M1, A1
	$18x - \dfrac{9}{2}x^{\frac{3}{2}} = 0 \Rightarrow x^{\frac{1}{2}} = (4)$	dM1
	$(16, 460.8)$	A1, A1
		(5)
(b)	$5 - \sqrt{x} = 0 \Rightarrow x = 25$	M1, A1
		(2)
(c)	$\int 9x^2 - \dfrac{9}{5}x^{\frac{5}{2}}dx = 3x^3 - \dfrac{18}{35}x^{\frac{7}{2}}$	M1A1
	Shaded area = $\underline{(25-16)\times 460.8} - \left[3x^3 - \dfrac{18}{35}x^{\frac{7}{2}}\right]_{16}^{25}$	<u>M1</u>, dM1
	$=1312.7$	A1
		(5)
		(12 marks)

10 (a)	Uses $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$ o.e. E.g. $\cos\theta\left(3\tan\theta + \dfrac{2}{\tan\theta}\right) \equiv \cos\theta\left(3\dfrac{\sin\theta}{\cos\theta} + \dfrac{2\cos\theta}{\sin\theta}\right)$	M1
	Uses $\sin^2\theta + \cos^2\theta = 1$ E.g. $\equiv 3\sin\theta + \dfrac{2\cos^2\theta}{\sin\theta} \equiv 3\sin\theta + \dfrac{2\left(1-\sin^2\theta\right)}{\sin\theta}$	dM1, A1
	$\equiv 3\sin\theta + \dfrac{2}{\sin\theta} - 2\sin\theta \equiv \sin\theta + \dfrac{2}{\sin\theta}$	A1*
		(4)
(b)	$\sin x + \dfrac{2}{\sin x} = 4\sin x - 5 \Rightarrow 3\sin^2 x - 5\sin x - 2 = 0$	M1, A1
	$\Rightarrow \sin x = 2, -\dfrac{1}{3} \Rightarrow x = 3.5$ for example.	dM1
	$\Rightarrow x = 3.48, 5.94$	A1
		(4)
		(8 marks)